AI/ML Internship – Task 4 Report

Classification Models Evaluation

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1. Introduction

Classification is one of the most widely used techniques in Machine Learning, where the goal is to predict categorical outcomes. In this task, we implemented a binary classification model using the Breast Cancer dataset to determine whether a tumor is malignant or benign.

This project focuses on building classification models, evaluating them using appropriate metrics, and handling class imbalance effectively. The models used in this task are Logistic Regression and Decision Tree Classifier.

2. Dataset Description

The dataset used is the Breast Cancer dataset available in scikit-learn. It contains various features computed from digitized images of breast mass.

- Total samples: 569
- Features: 30 numerical features
- Target classes:
    - 0 → Malignant (cancerous)
    - 1 → Benign (non-cancerous)

The dataset is slightly imbalanced, which makes evaluation beyond accuracy important.

3. Model Implementation

- Data Preprocessing
    - The dataset was split into training and testing sets using an 80:20 ratio.
    - Stratified sampling was used to preserve class distribution.
    - Feature scaling was applied using StandardScaler to normalize the data.

- Logistic Regression (Baseline Model)
  - A Logistic Regression model was trained as the baseline.
  - Maximum iterations were set to 1000 to ensure convergence.
  - Predictions were made on the test dataset.

- Decision Tree Classifier
  - A Decision Tree model was trained for comparison.
  - It helps in understanding non-linear relationships in data.
  - However, it is prone to overfitting.

4. Evaluation Metrics

- Confusion Matrix

The confusion matrix provides a detailed breakdown of predictions:

- True Positives (TP): Correctly predicted malignant cases
- True Negatives (TN): Correctly predicted benign cases
- False Positives (FP): Incorrectly predicted malignant
- False Negatives (FN): Missed malignant cases

[7]:
```
cm = confusion_matrix(y_test, y_pred)
print("Confusion Matrix:")
print(cm)

print("\nClassification Report:")
print(classification_report(y_test, y_pred))
```

```
Confusion Matrix:
[[41  1]
 [ 1 71]]

Classification Report:
              precision    recall  f1-score   support

           0       0.98      0.98      0.98        42
           1       0.99      0.99      0.99        72

    accuracy                           0.98       114
   macro avg       0.98      0.98      0.98       114
weighted avg       0.98      0.98      0.98       114
```

- Precision and Recall
  - Precision: Measures how many predicted positives are actually correct
  - Recall: Measures how many actual positives are correctly identified

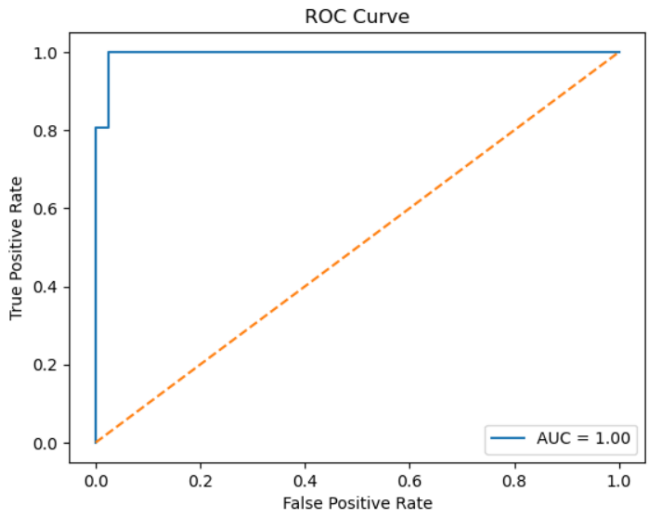
In medical diagnosis, Recall is more important because missing a cancer case can be dangerous.

- F1-Score
  - Harmonic mean of Precision and Recall
  - Useful when dealing with imbalanced datasets
  - Provides a balance between false positives and false negatives

- ROC Curve & AUC Score
  - ROC curve shows the trade-off between True Positive Rate and False Positive Rate
  - AUC (Area Under Curve) measures overall model performance
  - Higher AUC indicates better classification ability

```
auc = roc_auc_score(y_test, y_prob)

plt.plot(fpr, tpr, label=f"AUC = {auc:.2f}")
plt.plot([0, 1], [0, 1], linestyle='--')
plt.xlabel("False Positive Rate")
plt.ylabel("True Positive Rate")
plt.title("ROC Curve")
plt.legend()
plt.show()
```



5. Why Accuracy Alone is Insufficient

Accuracy can be misleading in imbalanced datasets. For example, if 90% of samples belong to one class, a model predicting only that class will still achieve 90% accuracy.

Thus, metrics like Precision, Recall, and F1-score are more reliable for evaluating model performance.

6. Handling Class Imbalance

To address class imbalance:

- Logistic Regression was trained using:

class_weight = "balanced"

- This assigns higher importance to the minority class.
- It improves the model's ability to correctly predict malignant cases.
- Helps in increasing Recall for critical predictions.

7. Model Comparison

Model	Strengths	Weaknesses
Logistic Regression	Stable, less overfitting	Assumes linearity
Decision Tree	Easy to interpret, flexible	Prone to overfitting

Observations:

- Logistic Regression provided consistent performance.
- Decision Tree showed signs of overfitting.
- Logistic Regression performed better in terms of generalization.

```
y_pred_tree = tree.predict(X_test)
print("\nDecision Tree Classification Report:")
print(classification_report(y_test, y_pred_tree))
```

```
Decision Tree Classification Report:
              precision    recall  f1-score   support

           0       0.85      0.93      0.89        42
           1       0.96      0.90      0.93        72

    accuracy                           0.91       114
   macro avg       0.90      0.92      0.91       114
weighted avg       0.92      0.91      0.91       114
```

```
[10]: from sklearn.metrics import accuracy_score

acc_lr = accuracy_score(y_test, y_pred)

acc_tree = accuracy_score(y_test, y_pred_tree)

comparison = pd.DataFrame({
    "Model": ["Logistic Regression", "Decision Tree"],
    "Accuracy": [acc_lr, acc_tree]
})

print(comparison)
```

```
                  Model  Accuracy
0  Logistic Regression  0.982456
1        Decision Tree  0.912281
```

8. Final Conclusion

- Logistic Regression is selected as the final model.
- It provides better stability and generalization.
- Evaluation metrics such as Recall and ROC-AUC confirm its effectiveness.
- Proper metric selection and imbalance handling significantly improve model performance.

9. Key Learnings

- Classification problems require careful metric selection.
- Accuracy alone is not reliable.
- Confusion matrix provides deeper insights into errors.
- Handling class imbalance is crucial in real-world datasets.
- Logistic Regression is effective for medical classification problems.